

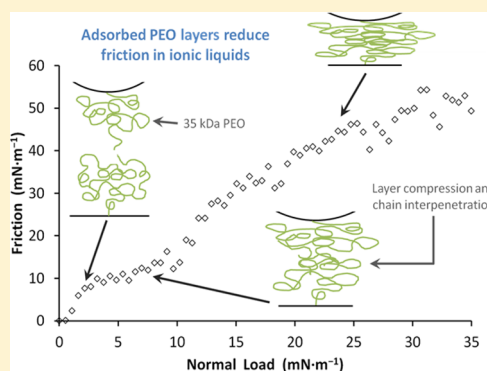
Poly(ethylene oxide) Mushrooms Adsorbed at Silica–Ionic Liquid Interfaces Reduce Friction

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S Supporting Information

ABSTRACT: The adsorbed layer conformation and lubricity of 35, 100, and 300 kDa PEO adsorbed to ionic liquid (IL)–silica interfaces from 0.01 wt % solutions have been investigated using colloid probe atomic force microscopy. The ILs used were propylammonium nitrate (PAN) and 1-butyl-3-methylimidazolium tetrafluoroborate ([BMIM][BF₄]), which are protic and aprotic ILs, respectively. Normal force curves reveal steric interactions consistent with adsorbed polymer layers which are best fit using the mushroom model. Friction measurements show that the adsorbed polymer layer markedly reduces friction compared to surfaces sliding in the pure ILs and that lubricity increases with polymer length. When polymer is adsorbed to the sliding surfaces, friction is controlled by the creation and disruption of intermolecular interactions between entangled chains and the dragging of polymer chains through the interpenetration region. These experiments show that added polymer can reduce friction while maintaining the useful properties of ILs as lubricants.



INTRODUCTION

Ionic liquids (ILs) are pure salts with melting points less than 100 °C.¹ The bulky ions employed in ILs frustrate packing arrangements and weaken Coulombic interactions compared to conventional salts, which drastically reduces the IL melting point compared to typical inorganic salts. Protic ILs (PILs) are synthesized by proton transfer from a Brønsted–Lowry acid to a Brønsted–Lowry base, while aprotic ILs (AILs) are synthesized via charge-transfer metathesis and quaternisation reactions. ILs have emerged as highly versatile solvents due to their favorable physicochemical properties, which generally include good electrical² and thermal³ stability, nonvolatility,³ and high thermal conductivity.⁴ ILs have been studied for use in a wide range of applications including as electrolytes in batteries⁵ and as catalysts.¹

ILs often self-assemble in the bulk⁶ and at interfaces^{7,8} into distinct nanostructures. Coulombic interactions and hydrogen bonding play important roles in driving IL nanostructure.⁶ The bulk and interfacial structures of the ILs studied here, the PIL propylammonium nitrate (PAN)^{9,10} and the AIL 1-butyl-3-methylimidazolium tetrafluoroborate ([BMIM][BF₄]),^{11,12} have been studied extensively. PAN self-assembles into an l₃-sponge phase in the bulk. The amphiphilic PA⁺ cations drive the solvophobic segregation of polar moieties from nonpolar moieties, forming distinct polar and nonpolar domains.¹⁰ Similar segregation of polar and nonpolar groups also occurs in [BMIM][BF₄]. Nonpolar domains consisting of (mainly butyl) alkyl chains are surrounded by a uniform distribution of polar groups consisting of the imidazole ring and the anion.¹²

IL physical properties such as low vapor pressure, high thermal stability, and conductivity, and their affinity for solid surfaces means they have attracted considerable research interest as lubricants in pure form^{13–15} and as lubricant additives to base oils with some striking results.^{13,16,17} Pure ILs are typically more effective lubricants than commonly used oils, and when ILs are combined with additives such as tricresyl phosphate¹⁸ and benzotriazole,¹⁹ even higher lubricity can be achieved. Similar increases in lubricity have been observed when *n*-alkanols²⁰ are added to PAN. Oil-soluble ILs have been investigated as additives to base oils. For example, a 1 mol % solution of a phosphonium IL in oil leads to a 5-fold reduction in friction compared to the pure oil, which is as effective as the pure IL.¹⁶

The conformation polymer chains adopt when adsorbed to a surface depends on the grafting density and relative affinities of the polymer for the surface and solvent.²¹ Polymers adsorbed as brushes have high grafting densities, such that the distance between grafting points is less than the mean end-to-end distance of the polymer in the melt (the Flory radius, *R_F*).²² Under these conditions the osmotic pressure within the polymer layer is high, and the polymers stretch to avoid contact with neighboring chains. This pressure balances the entropic cost of stretching beyond *R_F*; it is this balance that determines the equilibrium brush height and energy. Chains

Received: December 9, 2015

Revised: February 3, 2016



that are weakly attracted to the surface (low grafting densities) form mushrooms. For polymers adsorbed as mushrooms the layer height is less than R_F and the distance between chain grafting sites on the surface is greater than R_F .²² The osmotic pressure is lower in a mushroom than in a brush.

In previous work we have shown that for ethylammonium nitrate (EAN)–silica interfaces PEO²³ and several Pluronics²⁴ can compete effectively with the EA^+ cations and adsorb to silica. Colloid probe atomic force microscopy (AFM) force curves and fitting revealed that PEO adsorbs to the silica–EAN interface in a mushroom conformation.²³ This is in contrast to silica–aqueous interfaces where PEO adsorbs as a brush, consistent with water being a better solvent for PEO than EAN. EAN and PAN are structurally identical, except that the alkyl chain of PAN is one methylene group longer. As PEO is solvated in EAN and PAN via hydrogen bonds, and the bulk liquid hydrogen bond density is lower in PAN than EAN on account of the longer alkyl chain, PAN is a slightly poorer solvent for PEO than EAN. Rheology measurements have shown PAN is a theta-solvent for PEO, whereas EAN is a theta-good solvent.²⁵

Friction is markedly reduced when polymer brushes are adsorbed to solid surfaces in conventional solvents.²⁶ The loosely packed mushroom layer has a low osmotic pressure which enables chain interpenetration and entanglement during sliding, especially at high normal loads.²⁷ This leads to friction that is higher than for brush layers but still lower than for the bare surfaces as surface contact is prevented.

Dragging polymer chains through the interpenetration region is the main contribution to energy dissipation when two polymer bearing surfaces slide against each other.²⁸ For polymer brushes the interpenetration thickness, d , varies weakly (as $d \propto D^{-1/3}$) with surface separation, D , for moderate to high molecular weights up to moderate compressions. d increases more rapidly with D for low molecular weight polymer brushes²⁹ and for polymers adsorbed as mushrooms.²⁶ Shear thickening of the confined material may occur at high compressions, increasing friction.³⁰

MATERIALS AND METHODS

Chemicals. Propylammonium nitrate (PAN) was prepared by dropwise addition of equimolar quantities of nitric acid (RCI Labscan Ltd., Thailand, 70% w/w aqueous solution) to propylamine (Sigma-Aldrich, $\geq 99.0\%$ purity). The exothermic reaction was carried out in an ice bath which maintained the solutions at 8–10 °C to prevent the formation of oxide impurities. Excess water was removed by rotary evaporation (Buchi, Switzerland). Remaining water was removed by heating to 100–110 °C in an oil bath under a nitrogen atmosphere for 9 h. This resulted in a water content of less than 0.05 wt %, as determined by Karl Fischer titration. 1-Butyl-3-methylimidazolium tetrafluoroborate ([BMIM][BF₄], $\geq 97.0\%$ purity) was obtained from Sigma-Aldrich and used as received; the water content was measured to be 0.01 wt %.

PEO samples with molecular weights of 35, 100, and 300 kDa (Sigma-Aldrich) were used as received. Solutions of 0.01 wt % concentration of each molecular weight of PEO were prepared in PAN and [BMIM][BF₄]. PEO was dissolved by (separately) sonicating the samples and heating to 30–50 °C while stirring using a magnetic stirrer bar.

AFM Preparation. The flexural resonant frequencies and Q -factors of gold-coated (HQ:NSC36/tipless/Cr–Au, MikroMasch, Tallinn, Estonia) and uncoated (HQ:NSC36/tipless/No Al, MikroMasch, Tallinn, Estonia), tipless, n -type silicon cantilevers were measured using the cantilever tune function of the Tapping Mode Nanoscope AFM software. A colloid probe³¹ was prepared by gluing a 5 μ m

diameter borosilicate particle (Duke Scientific, Palo Alto, CA) onto each cantilever using a two part epoxy (Selley's Super Strength Araldite). After curing for at least 24 h cantilevers were imaged using an optical microscope (Zeiss, Axioskop 40) with built-in camera (Zeiss, AxioCam Cc1). Cantilever plane dimensions and the diameter of colloidal probes were measured using AxioVision software (Zeiss, AxioVision 4.7).

Normal spring constants can vary considerably, even for cantilevers fashioned from the same wafer.³² Normal and torsional cantilever spring constants were measured using the normal³³ and "hybrid"³⁴ Sader methods. The off-end loading correction³⁵ was applied to account for the fact that a normal load is not applied at the very end of the cantilever, but at a point a distance ΔL from its free end. ΔL is the distance that the center of the colloid probe is from the free end of the cantilever.

Prior to an experiment the AFM fluid cell, silicone O-ring, and other tubes and fittings were sonicated using an ultrasonic bath (Unisonics Australia Pty. Ltd.) in a solution of approximately 0.01 wt % sodium dodecyl sulfate for ~ 30 min, rinsed with reverse osmosis water and absolute ethanol, and dried in a laminar flow cabinet. The AFM cantilever was rinsed with reverse osmosis water and absolute ethanol and irradiated with UV/ozone radiation for 5–10 min to remove organic residues. A new colloid probe was used for each experiment to minimize contamination of the experimental system by polymer from previous experiments. The substrates for all experiments were (100) silicon wafers. The substrates were sonicated for 5 min in a 10 wt % solution of sodium hydroxide and then for 10 min in a 0.01 wt % solution of the surfactant hexadecyltrimethylammonium bromide. The substrates were then rinsed in Milli-Q water (18.2 M Ω -cm resistivity at 25 °C) and ethanol and then dried in a stream of nitrogen gas. Finally, the substrate was irradiated with UV/ozone radiation for 30 min to produce a highly hydrophilic surface.

The AFM fluid cell was assembled, and 0.5 mL of the pure solvent was injected. This was left to equilibrate for 30 min, before normal force curves were obtained to check that the system was polymer-free; the absence of long-range polymeric adhesion in force curves ensured this was the case. Friction measurements were performed following the acquisition of normal force curves.

After measurements in pure solvent 1 mL of PEO-IL solution was injected into the fluid cell, a five times replacement of the cell volume (total cell volume ≈ 0.2 mL),³⁶ which is sufficient for complete solution exchange. The system was then given 12 h to allow for polymer adsorption to both the silica surface and colloidal probe. The effect of adsorption time was studied by allowing the 300 kDa PEO 38 h to equilibrate. After this equilibration time nonadsorbed polymer was removed from the system by purging with 1 mL of pure IL. The system was left for a further 30 min to obtain thermal equilibrium before normal force and friction experiments were performed. For the purposes of this work "equilibration time" is the time the polymer remained in the AFM fluid cell before purging with the solvent.

AFM Force–Distance Profiles. Force–distance profiles were obtained using a Nanoscope IV MultiMode AFM (Bruker, Santa Barbara, CA) equipped with a PicoForce controller and scanner in contact mode. The closed loop feature of the PicoForce controller and scanner was disabled. Ramp sizes from 50 to 200 nm were used, with ramp rates from 0.015 to 1.0 Hz (corresponding to velocities from 3 to 50 nm s^{−1}). Force curves were obtained before friction measurements in the pure solvent and again before friction measurements in the polymer solution. Between normal force measurements the x - and/or y -offsets were incremented by at least 200 nm to prevent "memory"³⁷ effects arising from compression of the polymer layer. The normal force data were fit using a mushroom model and the Alexander–de Gennes (AdG) and Milner–Witten–Cates (MWC) models (see the Supporting Information for more details).

AFM Friction Measurements. Shear (friction) force as a function of normal load measurements were performed using the same AFM in friction force microscopy (FFM) mode. The closed loop feature of the PicoForce scanner and controller was disabled. The procedure for performing these measurements has been described in detail elsewhere.³⁸ For the experiments described in this study the scan

size and rate were set to $1\ \mu\text{m}$ and $5\ \text{Hz}$, respectively, corresponding to a scan velocity of $10\ \mu\text{m s}^{-1}$. The lateral voltage signal output by the AFM was converted to shear force using a custom written MATLAB 7.11.0 function.

Shear force was measured as a function of normal load for normalized loads from 0 to $35\ \text{mN m}^{-1}$. Normal and shear forces have both been normalized by the radius of the colloidal probe, R . For normal forces this gives a quantity proportional to the interaction free energy between two flat surfaces;³⁹ for shear forces it allows for direct comparison of shear forces to the normal forces in this study. The normal load was increased to its maximum value and then decreased until the tip became fully retracted. Each point on the shear force vs normal load plots represents an average of 16 forward (trace) and backward (retrace) motions of the colloidal probe over the same line on the surface. Friction coefficients were calculated by fitting a straight line to increasing load friction vs normal load data between normal loads of 15 and $35\ \text{mN m}^{-1}$. The error in the friction coefficients (given in Table 2) represents the standard deviation of the friction coefficients of three friction data sets for each molecular weight (and each equilibration time).

RESULTS AND DISCUSSION

Figure 1 shows force–distance profiles for silica–PAN interfaces with adsorbed 35, 100, and 300 kDa PEO. The

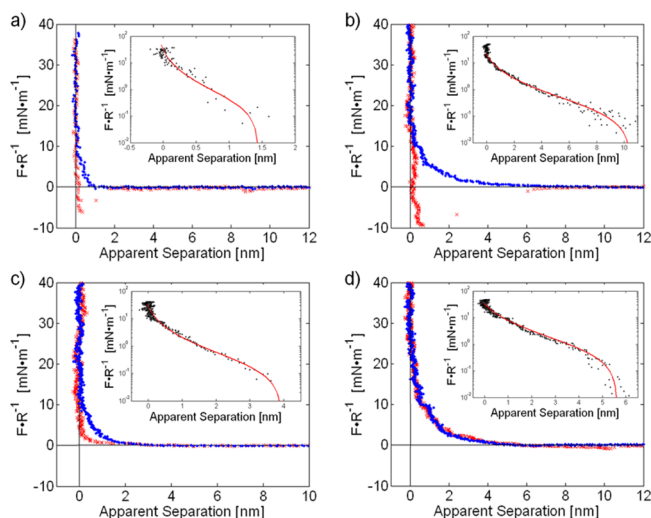


Figure 1. Force–distance profiles for a silica colloid probe with adsorbed PEO approaching (blue diamonds) and retracting from (red crosses) a planar silica surface with adsorbed PEO in PAN. The molecular weights of PEO adsorbed to each surface are (a) 35, (b) 100, (c) 300, and (d) 300 kDa (38 h equilibration time). The equilibration time is 12 h unless otherwise indicated. Insets show approach data plotted on log–linear axes with mushroom model fits (red solid line). Forces are normalized by the radius of the colloid probe, R .

steplike features observed for pure PAN–silica interfaces⁹ (a typical force–distance profile for pure PAN is given in Figure A of the Supporting Information) are not seen in any of the polymer systems. Rather, repulsive forces characteristic of steric interactions between adsorbed polymer layers are noted. This confirms that PEO adsorbs to the silica surfaces (probe and substrate), consistent with previous work for PEO adsorbed at EAN–silica interfaces.²³

As the surfaces approach, no force is measured at wide separations as the high PAN ionic strength screens long-range electrostatic and van der Waals interactions, at least to the sensitivity limit of the AFM. When the polymer layers adsorbed

to the colloid probe and the planar substrate make contact, steric interactions produce a repulsion.³⁷ The repulsive force increases as polymer layers are compressed until the cantilever spring constant is exceeded, and the force increases at zero apparent separation in the compliance region.²³ This means the thickness of the compressed polymer film cannot be ascertained using AFM. As polymer is adsorbed to both probe and the substrate, the distance which each polymer film is compressed is half the compression distance measured in AFM force–distance profiles.

The apparent separation at the onset of repulsion increases from 1 nm for 35 kDa PEO to 5 nm for 100 kDa PEO (cf. Figure 1). This is because the polymer extension into solution increases with molecular weight^{23,40,41} to reduce the osmotic pressure. To reduce osmotic pressure by the same amount, higher molecular weight polymer chains must stretch away from the surface more (on average) than lower molecular weight ones. The distances over which the polymers are compressed is considerably smaller than for the same PEO molecular weights in EAN (3.8 nm for 35 kDa PEO and 6.2 nm for 100 kDa PEO)²³ because PAN is a poorer solvent for PEO than EAN; the polymer thus adopts a flatter, more compressed conformation in PAN.

Fits to the force data were attempted using the mushroom model^{42,43} and the brush models of Alexander–de Gennes (AdG)⁴³ and Milner–Witten–Cates (MWC).⁴⁴ Acceptable fits could only be obtained using the mushroom model. These are shown in the log–linear insets to Figure 1, and the associated fit parameters are in Table 1. This finding is consistent with

Table 1. Parameters Used To Generate the Fits Displayed in the Insets to Figure 1, Using the Mushroom Model^{42 a}

PEO mol wt (kDa)	anchoring point distance (nm)	equilibrium layer thickness (nm)	incompressible layer thickness (nm)
35	5.2	1.8	1.1
100	7.2	5.9	0.6
300	6.8	2.7	0.8
300 (38 h)	3.8	3.9	1.1

^aThe equilibration time is 12 h unless otherwise indicated.

PEO adsorption from EAN onto silica, which was also found to be best approximated by a mushroom conformation.²³ The fit attempts using the AdG and MWC models are presented Figure D, and the fit parameters are in Tables A and B of the Supporting Information. The quality of these fits can be assessed qualitatively by visual inspection or quantitatively by using eq S4. Table F of the Supporting Information shows the results of this error analysis to assess the quality of the mushroom, AdG, and MWC fits to the experimental data. The mushroom model gives a quantitative fit that is demonstrably better for most systems. This indicates the PEO tends to adsorb at the silica–PAN interface in a mushroom morphology.

The repulsive force measured when polymer layers are compressed is a consequence of an increase in osmotic pressure and a decrease in elastic energy.⁴³ Both factors contribute additive terms to the AdG and MWC models, and it is the balance of these factors that largely determines the force profiles between brush bearing surfaces. When two mushroom bearing surfaces approach, the greater distance between chain anchoring points means the mushrooms have space to flatten and expand parallel to the surface.²³ The osmotic pressure is thus lower in a mushroom than a brush, and the main

contribution to the force profile is the decrease of elastic energy upon compression.

The compression distance for 300 kDa PEO after 12 h equilibration is 3.5 nm (Figure 1c). For the same equilibration time, the compression distance of 100 kDa PEO is 6 nm despite the shorter chain. This result is a consequence of the slower dynamics of long polymer chains compared to short ones; when the equilibration time for 300 kDa PEO was increased to 38 h (Figure 1d), the compression distance increased to ~ 6 nm. The effect of the longer equilibration time is twofold: it allows more polymer chains to adsorb and gives more time for the polymer to transition from a flat, compact layer to a more extended conformation. Slow structural changes upon polymer adsorption are common due to entropic considerations,^{43,45} and the high viscosity of the polymer/PAN mixtures means that relaxation times are much longer in PAN than in water. Both the adsorption of additional polymer chains and their conformational changes are reflected by the distance between chain anchoring points decreasing and the equilibrium brush layer thickness increasing (cf. Table 1). These effects caused a slight increase in the quality of the AdG and MWC fits to the data recorded after a 38 h equilibration time (cf. Figure D of the Supporting Information), but the mushroom model fit is still superior. It is likely that even longer equilibration times would lead to more extended PEO layers, but instrumental considerations meant that longer time periods could not be probed here.

Adhesion is noted in the retraction data in Figure 1 for PEO molecular weights of 35 and 100 kDa, but not for 300 kDa. Adhesion is caused by interpenetration and subsequent entanglement of polymer chains on opposing surfaces and/or the formation of attractive interactions (e.g., van der Waals (vdW) forces and H-bonds) between overlapping chains. The critical average molecular weight above which entanglement between PEO chains may occur in the bulk is ~ 10 kDa;⁴⁶ thus, all polymers studied here may entangle. Loosely packed layers interpenetrate more easily and are more prone to adhesion than densely packed layers. From Table 1, the distance between anchoring sites in 300 kDa PEO decreases compared to 100 kDa, meaning that the density of polymer in the adsorbed layer increases for the higher molecular weight. Consequently, adhesion was not observed in the 300 kDa PEO system because the greater polymer layer density prevents interpenetration.

Figure 2 shows force–distance profiles measured between two silica surfaces following adsorption of PEO from [BMIM][BF₄]. The structural forces⁴⁷ present in pure [BMIM][BF₄] (see Supporting Information, Figure B) are replaced by steric repulsion when PEO is present. For all molecular weights the compression distances in [BMIM][BF₄] are small (≤ 2 nm), suggesting the polymer adopts a flat and/or incompressible conformation on the surface. There is a weak dependence of compression distance on molecular weight; 0.5 nm for 35 kDa increasing to 2 nm for 300 kDa. Increasing the equilibration time for 300 kDa PEO in [BMIM][BF₄] from 12 to 37 h (Supporting Information Figure C) had minimal effect on the compression distance, suggesting the polymer chains do not change conformation.

The log–linear insets in Figure 2 show approach data fitted using the mushroom model. Best fit attempts using the mushroom, MWC and AdG models are given in the Supporting Information Figure E and fitting parameters in Tables C–E. The results of using eq S4 to quantitatively assess the quality of

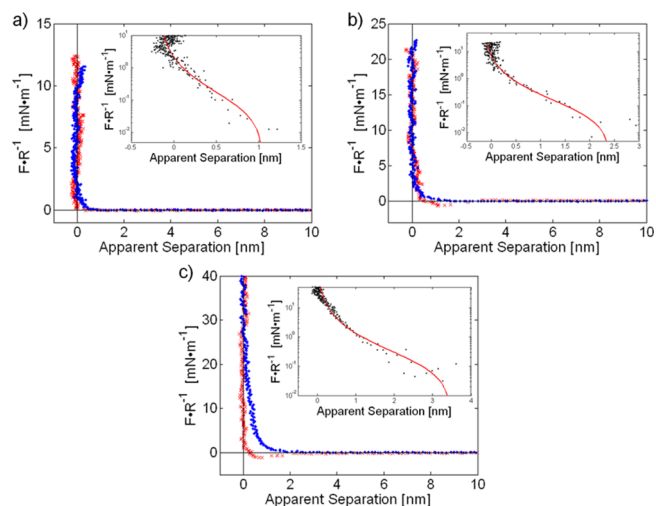


Figure 2. Force–distance profiles for a silica colloid probe with adsorbed PEO approaching (blue diamonds) and retracting from (red crosses) a planar silica surface with adsorbed PEO in [BMIM][BF₄]. The molecular weights of PEO adsorbed to each surface are (a) 35, (b) 100, and (c) 300 kDa. The equilibration time for each system is 12 h. Insets show approach data plotted on log–linear axes with mushroom model fits (red solid line). Forces are normalized by the radius of the colloid probe, R .

the fits are given in Table G of the Supporting Information. As with PAN, this analysis indicates the fit of the mushroom model is as good, or significantly better, than either the MWC or AdG models.

Comparison of results in Figures 1 and 2 indicates that PEO chains are more compressible in PAN than [BMIM][BF₄]. At first glance this is surprising given that previous SANS experiments⁴⁸ showed [BMIM][BF₄] is a good solvent for PEO, whereas PAN is a theta solvent.²⁵ Polymer adsorption to a surface is a complex function of polymer–surface, solvent–surface, polymer–solvent, and polymer–polymer interactions. Polymer–surface interactions, as well as competition between polymers and solvent ions or molecules, can change the conformation of a polymer when adsorbed at an interface compared to the bulk conformation. The primary ammonium cation of PAN adsorbs more strongly to silica charge sites than the sterically hindered, delocalized imidazolium cations,⁹ which means that PEO can compete more effectively for adsorption sites in [BMIM][BF₄] than PAN, resulting in a flatter conformation.

Figure 3 shows friction as a function of normal load for the PAN/PEO (Figure 3a) and [BMIM][BF₄]/PEO (Figure 3b) systems. To obtain these data, the colloid probe traced over the same line on the surface many times at different normal loads. It is worth noting that the friction remained low up to the maximum normal load investigated, confirming the polymer remained adsorbed to the silica surface and did not undergo any catastrophic conformational rearrangements. No significant differences were observed when the normal load was decreased so for clarity only the increasing load data are presented.

The friction data for pure PAN in Figure 3a are consistent with previous results on mica.^{20,38,49} An increase in friction from a normalized shear force of 0–20 mN m^{−1} occurs at a normal load of 2.5 mN m^{−1}. Similarly in pure [BMIM][BF₄] there is an increase in friction from 0 to 10 mN m^{−1} at 2 mN m^{−1} normal load. These friction increases are caused by removal of near surface ion layers from between the colloid

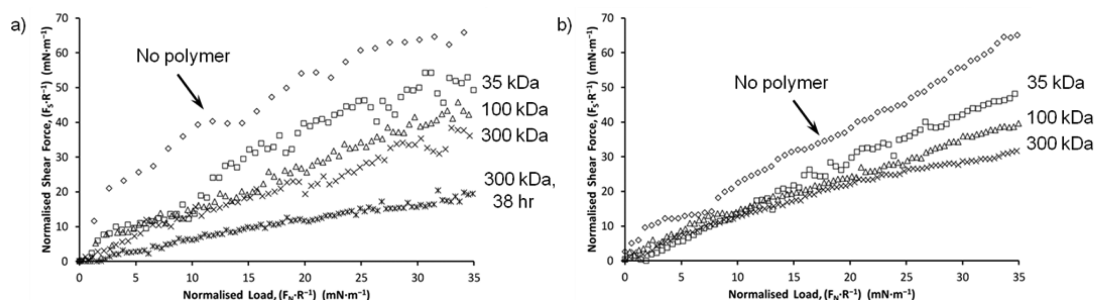


Figure 3. Normalized shear force as a function of normalized load for a silica colloid probe sliding against a planar silica surface in (a) PAN and (b) [BMIM][BF₄]. Data are presented for both bare silica surfaces (labeled “no polymer”) and for silica surfaces with adsorbed PEO (labeled with the polymer molecular weight). The equilibration time is 12 h unless otherwise indicated. All forces are normalized by the radius of the colloid probe, R . A sliding velocity of $10 \mu\text{m s}^{-1}$ was used for all data.

Table 2. Friction Coefficients (μ) for a Silica Colloid Probe Sliding against a Planar Silica Surface in PAN and [BMIM][BF₄]^a

IL	μ , no polymer	PEO mol wt (kDa)			
		35	100	300	300 (38 h)
PAN	1.02 ± 0.23	1.26 ± 0.18	1.15 ± 0.08	1.06 ± 0.11	0.52 ± 0.07
[BMIM][BF ₄]	1.57 ± 0.21	1.29 ± 0.09	1.02 ± 0.15	0.72 ± 0.16	N/A

^aData are presented for both bare silica surfaces (labeled “ μ , no polymer”) and for silica surfaces with adsorbed PEO (labeled with the polymer molecular weight). The equilibration time is 12 h unless otherwise indicated.

probe and the surface as load is increased. For this to happen attractive interactions, especially solvophobic interactions and vdW forces, between near-surface ions and ions adsorbed to the surface must be disrupted, which results in energy dissipation and friction.³⁸ Once near surface ion layers are displaced, the boundary layer separates the sliding surfaces and friction increases more slowly. There is a second, smaller increase in friction in pure [BMIM][BF₄] at a normal load of 8 mN m^{-1} . This is probably the result of the rearrangement of surface adsorbed ions as pressure increases. Similar effects have been observed previously in other pure IL systems.¹⁵

Adsorption of polymer at the interfaces suppresses the sudden increase in friction noted for the pure ILs by inhibiting layering of IL ions. Friction increases with increasing normal load and decreases with increasing polymer molecular weight for a given load. This is consistent with previous reports for the effect of polymer molecular weight on friction in both aqueous⁵⁰ and nonaqueous^{40,41} solvents.

Friction data for both IL/PEO systems shows small (ca. 13%) progressive reductions in friction with each tripling of the molecular weight for a given equilibration time. This is a consequence of higher attractive interchain forces (e.g., vdW) with increasing polymer molecular weight. Stronger intermolecular interactions reduce both the amount of interpenetration by polymer chains on opposing surfaces^{26–28} and the number of conformation changes that take place during sliding.^{38,51,52} Interpenetration and conformation changes are the dominant dissipation pathways available to sliding polymer bearing contacts so reducing their frequency of occurrence necessarily reduces friction. Confinement can also increase the polymer layer viscosity,³⁰ and higher polymer molecular weights produce larger viscosity increases for a given compression.³⁰ This could also contribute to lower friction for higher molecular weight PEO, as high viscosity suppresses conformational changes. As PEO polymer layers are incompressible in [BMIM][BF₄] regardless of molecular weight, this means the reduction in friction with molecular weight is likely due to more

rigid polymer layers which will have fewer degrees of freedom for energy dissipation.⁵³

Friction was 50% lower for 300 kDa PEO/PAN after 38 h adsorption (cf. Figure 3a) compared to 12 h. This is consistent with more extended layer conformations forming over longer time periods (cf. Figure 1). Friction vs normal load data for 300 kDa PEO/[BMIM][BF₄] with equilibration times of 13 and 37 h are given in Figure F of the Supporting Information. As with the normal force–distance data, there was no significant effect of equilibration time on friction for the [BMIM][BF₄]/PEO system.

Table 2 shows the friction coefficients for both IL/PEO systems. The friction coefficient for neat PAN is broadly consistent with previous studies on mica,^{20,38} with the small variation likely due to silica cleaning protocols.⁵⁴ In pure PILs at high normal load energy is dissipated during sliding via ion conformation changes and structural deformations (e.g., gauche defects, chain tilting, etc.),³⁸ and the friction coefficient provides a measure of the relative amount of energy dissipated from these conformation changes.

The friction coefficient for neat [BMIM][BF₄] is much greater than a previous microtribology study ($\mu \approx 0.05$)⁵⁵ involving diamond-like carbon and boron carbon nitride surfaces. This is due to the different geometries involved in microscale ball-on-disk tests compared to nanoscale AFM experiments, differences in scale and therefore lubrication mechanism,⁵⁶ and differences in surface roughness.

Table 2 shows a weak trend for PAN/PEO systems of decreasing friction coefficient with increasing polymer molecular weight. This is consistent with the polymer layer becoming more extended as molecular weight increases which is a consequence of the higher osmotic pressures that are generated for higher polymer molecular weights.³⁷ Chain interpenetration is low in highly stretched polymer layers due to strong steric interactions between layers,⁵⁷ and thus the contribution to friction caused by dragging interdigitated polymer chains is reduced. The friction coefficient for the 300 kDa system decreases markedly as equilibration time increased from 12 to

38 h, which is similarly consistent with the extension of adsorbed polymers into solution that occurs with time in these systems. Figure 3a shows a gradual increase in the friction coefficient of PAN/35 kDa PEO from 0.80 to 1.08 occurs at 10–13 mN m⁻¹ normal load. A more subtle increase occurs in 100 kDa PEO at 10 mN m⁻¹ normal load. This corresponds to the load in the normal force curves where the polymer layers become incompressible. Similar behavior in poly(L-lysine)-graft-dextran has been attributed to penetration of the polymer layer on one surface by the layer on the opposing surface once the normal load exceeded some critical, polymer conformation-dependent normal load.²⁶ H-bonds can form between overlapping chains and entanglement can occur, provided polymer molecular weight is sufficiently large.⁵⁸ Entanglement is consistent with the observation of adhesion in force–distance profiles (cf. Figure 1a,b) for these polymer molecular weights. No change in the friction coefficient of the 300 kDa PEO system is observed, which is further evidence the adsorbed layer is more dense at this molecular weight. This is consistent with the absence of adhesion in force–distance profiles for this PEO molecular weight (cf. Figure 1c,d).

Table 2 shows that the friction coefficient of 35 kDa PAN/PEO is almost within error of 100 kDa PAN/PEO. This result suggests that there is comparable interpenetration of polymer chains in the 100 and 35 kDa systems. The increase in chain length as polymer molecular weight increases has two competing effects. Based on the fact that for any polymer there exists a minimum chain length for interpenetration,^{46,59} long chains are more likely to interpenetrate than short ones. However, the osmotic pressure between polymer layers, which increases with polymer molecular weight, acts to reduce polymer chain interpenetration.³⁷ To explain the similarity in the friction coefficients, the second effect is only slightly more dominant as molecular weight increases from 35 to 100 kDa in the PAN/PEO system. The adhesion force is larger in 100 kDa PEO than in 35 kDa PEO (cf. Figure 1a,b) which suggests a greater amount of chain interpenetration consistent with the first effect. Furthermore, the increase in the compression distance (4 nm increase) in force–distance profiles as molecular weight increases from 35 to 100 kDa is not large, suggesting that the osmotic pressure does not increase significantly during this transition. These two results suggest that the amount of extension brought about by the change in conformation as PEO molecular weight increases from 35 to 100 kDa only just increases osmotic pressure enough to counter the increase in interpenetration enabled by the increase in chain length.

Similar to the overall trend in PAN/PEO systems, friction coefficients decrease with increasing polymer molecular weight in [BMIM][BF₄]/PEO systems. Apart from an increase in the force required to compress higher molecular weight polymer layers to the hard wall repulsion, force–distance profiles show very little change as polymer molecular weight increases from 35 to 300 kDa in [BMIM][BF₄] (cf. Figure 2). This increase in force required for compression is consistent with PEO layers becoming more densely packed as molecular weight increases. The mass of adsorbed polymer generally increases with molecular weight.⁶⁰ To facilitate this, packing density is expected to increase resulting in the layer becoming less compliant and increasing the compression forces observed in force–distance profiles, consistent with the results in Figure 2. Higher packing densities promote attractive interactions between polymer chains on the same surface, reducing

interpenetration between polymer chains and therefore friction. That the friction coefficient decreases with polymer molecular weight in the [BMIM][BF₄]/PEO system is consistent with dense layers forming and little chain interpenetration.

CONCLUSIONS

Poly(ethylene oxide) (PEO) adsorbs onto silica from both propylammonium nitrate (PAN) and 1-butyl-3-methylimidazolium tetrafluoroborate ([BMIM][BF₄]) in a relatively flat “mushroom” conformation. When polymer is added to either PAN or [BMIM][BF₄], the interfacial layering of IL ions is disrupted as polymer chains replace IL ions in the vicinity of the solid surface. Friction is reduced compared to the pure IL, with larger polymer molecular weights producing lower friction. In PAN, favorable polymer–solvent interactions cause PEO to adopt a more elongated mushroom conformation as polymer molecular weight increases. Similarly, increasing the adsorption times for high molecular weight PEO results in the polymer layer becoming slightly more extended, which indicates the equilibrium may not yet have been reached. Conversely, the conformation of PEO adsorbed at the silica/[BMIM][BF₄] interface does not change with polymer molecular weight or equilibration time.

Interpenetration of polymer chains adsorbed to opposing silica surfaces occurs for 35 and 100 kDa PEO in PAN, but not for 300 kDa in PAN or any molecular weight PEO in [BMIM][BF₄]. Interpenetration of polymer chains leads to chain entanglement which facilitates attractive interactions between PEO chains adsorbed to opposing surfaces. These adhesive interactions lead to greater energy dissipation by polymer chains as they are dragged through the region of interpenetration and thus higher friction. No interpenetration occurs when the density of the adsorbed layers is sufficiently high, reducing friction.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.langmuir.5b04503.

Force–distance profiles for pure PAN and [BMIM][BF₄], force–distance profiles and friction data for [BMIM][BF₄] as a function of equilibration time, formulas for the mushroom, Milner–Witten–Cates (MWC) and Alexander–de Gennes (AdG) models, fits to force–distance data using these models, the parameters used for the fits, and a quantitative calculation of the errors associated with these fits (PDF)

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Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

R.A. thanks the Australian Research Council Future Fellowship (FT120100313). This research was supported by an Australian Research Council Discovery Project (DP120102708) and Equipment Grant (LE110100235). J.S. thanks the University of Newcastle, Australia, for the provision of a PhD (APA) scholarship.

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